Math 150: Math Modeling

Modeling the American Workforce: A Compartmental Analysis of U.S. Employment Status (1955-Present)

By: Alexzia Gutierrez, Tina Li, Sharlize Berrospe, Kurt Qin, Yi Fei Tao

Background

Why do we care?

- ★ Important signal of changes and current conditions in the labor market and economy.
- ★ Measures economic health for expansions and recessions
- ★ Determines interest rates and monetary policies for stable pricing

What are we observing?

- ★ Long-term labor trends in the U.S. from 1955 to present to understand how the workforce grows and responds to economic shocks
- ★ This study identifies how employment levels deviate from their long-term linear growth during periods of instability

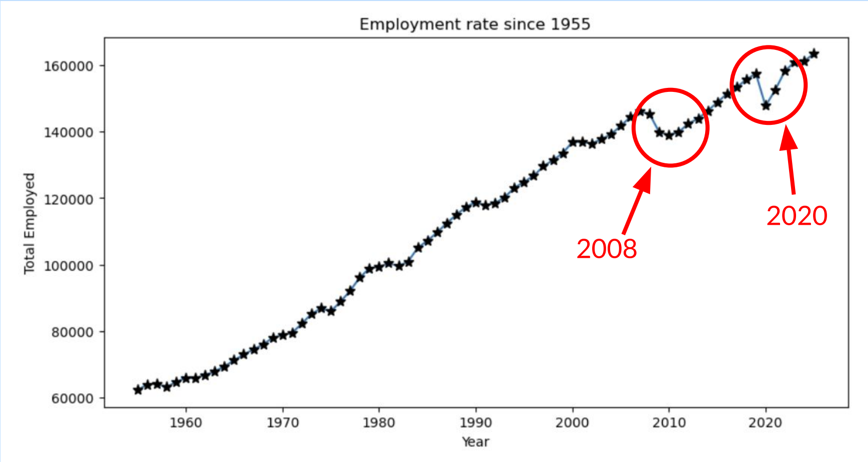
Origin:

- ★ U.S. Bureau of Labor Statistics (BLS)
- ★ Collected via the Current Population Survey (CPS)
    - ○ Tracks the civilian noninstitutional population

Data

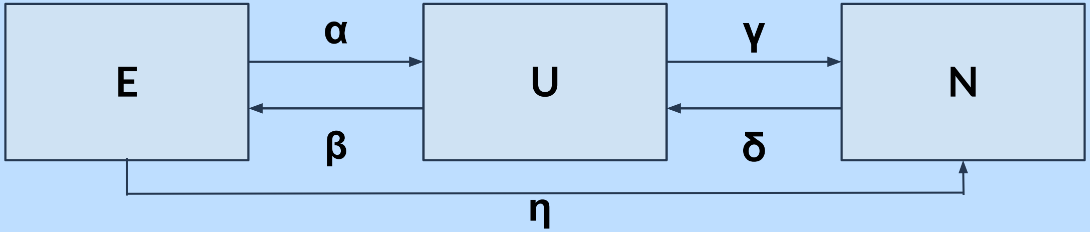
★ **Observed trends:**

- Overall linear increase in employment
- Dips in total employment rate (E) coincide with major U.S. recessions:
  - 2020 Pandemic Recession (quick recovery)
  - 2008 Great Recession (very slow recovery)
  - Recessions in 1980-1982, 1990-1991, 1973-1975
- Employment rate is a "lagging" indicator, so it drops after a recession begins and recovers much later than when the economy starts to grow again.
- Different recessions shift people differently between compartments (*E*→*U*, *E*→*N*)



➔ Employed civilians (**E**)
➔ Unemployed civilians in the labor force (**U**)
➔ Unemployed civilians out of the labor force (**N**)

Model



Employed (E)

- ★ people who currently have a job

Unemployed (U)

- ★ people who are jobless, but are actively looking for work in the last 4 weeks

Not in labor force (N)

- ★ people who are not working and not seeking employment

Total population (M)

- ★ civilian noninstitutional population age 16 and older that can be employed

Time (t)

Parameters

α - rate employed people lose their job
β - rate unemployed people find jobs
γ - rate unemployed people stop searching and leave labor force
δ - rate people not in the labor force start searching for jobs
η - rate employed people leave the labor force entirely

Units

$[\alpha] = [\beta] = [\gamma] = [\delta] = [\eta] = 1/\text{year}$
$[t] = \text{year}$
$[E] = [U] = [N] = \text{people}$, $[M] = \text{people}$

System of Differential Equations

$dE/dt = \beta U - \alpha E - \eta E$
$dU/dt = \alpha E + \delta N - \beta U - \gamma U$
$dN/dt = \eta E + \gamma U - \delta N$

Assumptions

- ★ Constant total population, M
- ★ Constant parameters
- ★ People in each compartment has the same transition rate

Nondimensionalization

Independent variable: $[t] = year,\ [\beta] = \frac{1}{year}$

$\implies [t][\beta] = year \times \frac{1}{year} = 1,\ \frac{[t]}{[1/\beta]} = \frac{year}{year} = 1$

$$t^* = \frac{t}{1/\beta} = \beta t \to t = \frac{t^*}{\beta}$$

Dependent variables: $[E] = [U] = [N] = people, [M] = people$

$\implies \frac{[E]}{[M]} = \frac{[U]}{[M]} = \frac{[N]}{[M]} = \frac{people}{people} = 1$

$$E^* = \frac{E}{M} \to E = E^* M$$

$$U^* = \frac{U}{M} \to U = U^* M$$

$$N^* = \frac{N}{M} \to N = N^* M$$

β: the rate at which unemployed individuals find jobs (U→E)

1/β: The time spent on finding a job

t*: The proportion of time(t) to the time spent on job hunting(1/β)

E*: The proportion of people currently have a job(E) in the total population(M).

U*: The proportion of people currently unemployed and are looking for a job(U) in the total population(M).

N*: The proportion of people currently don't have a job(E) and do not look for a job in the total population(M).

Nondimensionalization

$$\frac{dE^*}{dt^*} = U^* - R_0E^* - R_1E^*$$

$$\frac{dU^*}{dt^*} = R_0E^* + R_2N^* - U^* + R_3U^*$$

$$\frac{dN^*}{dt^*} = R_1E^* + R_3U^* - R_2N^*$$

- ★ The nondimensional system describes how the proportions of employed, unemployed, and not in labor force evolve over scaled time.
- ★ Each equation represents the balance of inflows and outflows between compartments, governed by dimensionless parameters that capture the relative strengths of transition rates.

$$\frac{dE}{dt} = \beta U - \alpha E - \eta E$$

$$\rightarrow \frac{d(E^*M)}{d(\frac{t*}{\beta})} = \beta(U^*M) - \alpha(E^*M) - \eta(E^*M)$$

$$\rightarrow (\beta M)\frac{dE^*}{dt^*} = (\beta M)U^* - (\alpha M)E^* - (\eta M)E^*$$

$$\rightarrow \frac{dE^*}{dt^*} = U^* - \frac{\alpha}{\beta}E^* - \frac{\eta}{\beta}E^*$$

Set $\frac{\alpha}{\beta} = R_0, \frac{\eta}{\beta} = R_1$

$$\Rightarrow \frac{dE^*}{dt^*} = U^* - R_0E^* - R_1E^*$$

$$\frac{dU}{dt} = \alpha E + \delta N - \beta U - \gamma U$$

$$\rightarrow \frac{d(U^*M)}{d(\frac{t*}{\beta})} = \alpha(E^*M) + \delta(N^*M) - \beta(U^*M) - \gamma(U^*M)$$

$$\rightarrow (\beta M)\frac{dU^*}{dt^*} = (\alpha M)E^* + (\delta M)N^* - (\beta M)U^* - \gamma MU^*$$

$$\rightarrow \frac{dU^*}{dt^*} = \frac{\alpha}{\beta}E^* + \frac{\delta}{\beta}N^* - U^* + \frac{\gamma}{\beta}U^*$$

Set $\frac{\delta}{\beta} = R_2, \frac{\gamma}{\beta} = R_3$

$$\Rightarrow \frac{dU^*}{dt^*} = R_0E^* + R_2N^* - U^* + R_3U^*$$

$$\frac{dN}{dt} = \eta E + \gamma U - \delta N$$

$$\rightarrow \frac{d(N^*M)}{d(\frac{t*}{\beta})} = \eta(E^*M) + \gamma(U^*M) - \delta(N^*M)$$

$$\rightarrow (\beta M)\frac{dN^*}{dt^*} = (\eta M)E^* + (\gamma M)U^* - (\delta M)N^*$$

$$\rightarrow \frac{dN^*}{dt^*} = \frac{\eta}{\beta}E^* + \frac{\gamma}{\beta}U^* - \frac{\delta}{\beta}N^*$$

$$\Rightarrow \frac{dN^*}{dt^*} = R_1E^* + R_3U^* - R_2N^*$$

Steady State

(non-dimensionally):

Conditions:

$$\frac{dE^*}{dt^*} = U^* - (R_0 + R_1)E^* = 0$$

$$\frac{dU^*}{dt^*} = R_0E^* + R_2N^* - (1 - R_3)U^* = 0$$

$$\frac{dN^*}{dt^*} = R_1E^* + R_3U^* - R_2N^* = 0$$

Constraint: $E^* + U^* + N^* = 1$

Final solution:

$$E^* = \frac{1}{1 + (R_0 + R_1) + \frac{1}{R_2}\left(R_1 + R_3(R_0 + R_1)\right)}$$

$$U^* = (R_0 + R_1)E^*$$

$$N^* = \frac{E^*}{R_2}\left(R_1 + R_3(R_0 + R_1)\right)$$

(dimensionally):

Conditions:

$$\frac{dE}{dt} = \beta U - \alpha E - \eta E = 0$$

$$\frac{dU}{dt} = \alpha E + \delta N - \beta U - \gamma U = 0$$

$$\frac{dN}{dt} = \eta E + \gamma U - \delta N = 0$$

Solve the System: $\begin{cases} \beta U - \beta(R_0 + R_1)E = 0 \\ \beta R_0 E + \beta R_2 N - \beta(1 + R_3)U = 0 \\ \beta R_1 E + \beta R_3 U - \beta R_2 N = 0 \end{cases}$

Total population constraint: $E + U + N = M$

Final solution:

$$E = \frac{M}{1 + (R_0 + R_1) + \frac{1}{R_2}\left(R_1 + R_3(R_0 + R_1)\right)}$$

$$U = M(R_0 + R_1)E$$

$$N = M\frac{E}{R_2}\left(R_1 + R_3(R_0 + R_1)\right)$$

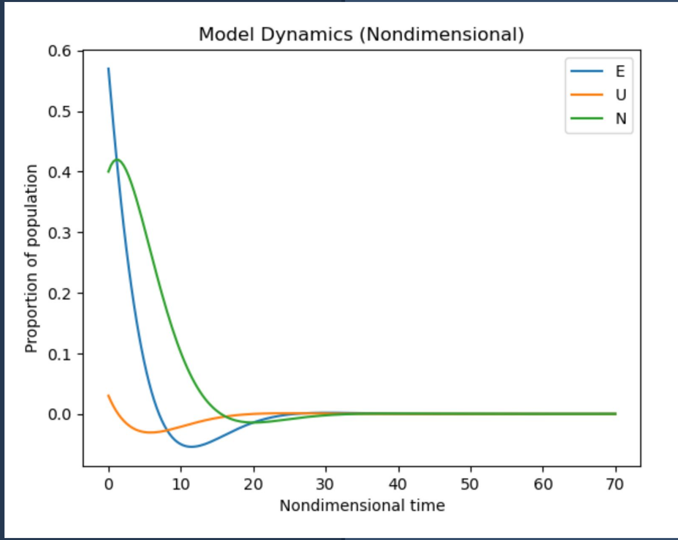
Model Solution

Initial Conditions and Parameters

For our model we used the following initial conditions :

- ➔ A starting population of 100,000
- ➔ $\alpha = 0.04$
- ➔ $\beta = 0.30$
- ➔ $\gamma = 0.08$
- ➔ $\delta = 0.05$
- ➔ $\eta = 0.05$
- ➔ $E[0] = 0.57$
- ➔ $U[0] = 0.03$
- ➔ $N[0] = 0.4$

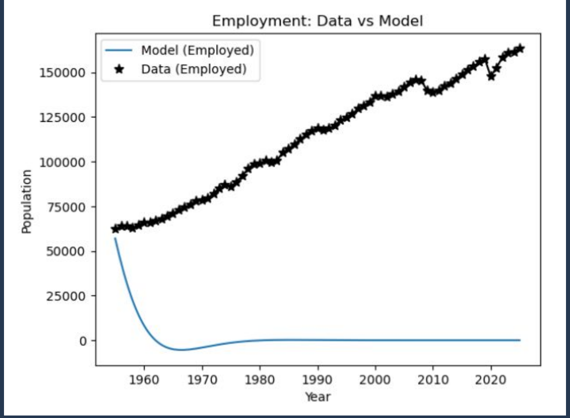
We also made the assumption our population and transition rates were fixed.

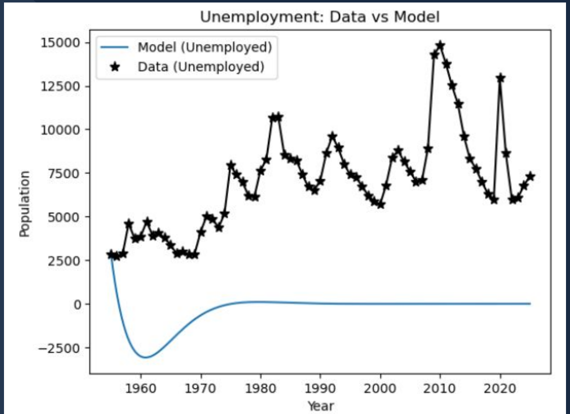


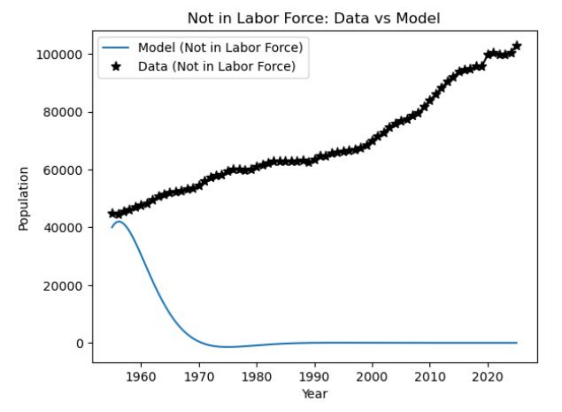
State Behavior

- ★ All three compartments experience rapid change initially
- ★ System converges to a stable equilibrium determined by transition rates
- ★ Variables are strongly coupled, so changes in one compartment affect the others
- ★ The system shows smooth, exponential-type behavior
- ★ All compartments decrease over time and approach zero

Comparing Model Solution with Data







- ★ Models converge to zero population; Not conserved.
- ★ Negative population from unemployed likely caused population loss.
- ★ Potential improvements:
  - ○ 2nd Derivative ODEs, to specify I.C.s at end of time.
  - ○ Fourth ODE for source population, increasing over time.

Conclusion

- ★ For future modeling, attempt different systems of ODEs, focused on incorporating changing population over time.
- ★ Following SIS model appeared to work well on paper, in program not so much.
- ★ Hardest challenge was to address variable population.
- ★ Analyzing employment statistics in different time periods was the most interesting, e.g. 2008, 2020.